Evidential Reliable Fusion for Partial Multi-View Incomplete Multi-Label Classification

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***Abstract*—In recent years, the missing data problem in multi-view multi-label classification (MvMlC) has attracted extensive attention from researchers, with numerous solutions for partial multi-view incomplete multi-label classification (PMvIMlC) emerging. Nevertheless, two critical challenges persist. One is sub-optimal coarse-grained multi-view fusion: traditional dynamic fusion at the view level is unable to accommodate the practical fusion demands of samples with diverse qualities. The other is neglecting latent information within missing labels: during the training phase, existing works only focus on the limited supervised information of unmissing labels while ignoring the underlying information at missing positions. To address these issues, we propose Evidential Reliable Fusion for Partial Multi-view Incomplete Multi-label Classification, termed ERF. ERF comprises two core modules: 1) Uncertainty-guided fusion module via evidence theory and 2) adaptive negative label pseudo-labeling. The former quantifies sample-level uncertainty of each view based on evidence theory, which is then used to guide multi-view fusion, enabling a fine-grained, instance-level multi-view fusion scheme. For the latter, leveraging the model's perception ability for neighboring samples in the label space, we design a strategy to select reliable negative pseudo-labels. This module enhances supervisory information to aid model training by recovering reliable negative pseudo-labels. Extensive experiments demonstrate that our ERF delivers significantly superior classification performance over existing methods.**

***Index Terms*—Partial multi-view learning, incomplete multi-label learning, multi-label classification, multi-view uncertainty, evidential learning.**

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I. Introduction

MULTI-VIEW multi-label classification (MvMlC) is a composite task that integrates multi-view learning and multi-label learning. In particular, multi-view learning refers to a learning paradigm that deals with heterogeneous data representing different representations of the same subject [1], [2], [3], [4], [5], with the objective of addressing downstream tasks through information fusion of complementary views. Given its competence in integrating complementary information, multi-view research has been extensively utilized across numerous fields, including medical diagnostics [6], action recognition [7], stock ranking [8], and recommendation systems [9], [10]. Correspondingly, multi-label classification describes frameworks associating multiple labels with individual instances concurrently. For example, a landscape image may be described using tags such as "sky", "mountain", and "house". Similarly, a news article might be characterized by labels like "sports", "football", and "match"[11], [12]. Given that real-world objects inherently possess multiple co-occurring attributes, multi-label learning offers significant practical applicability. Consequently, MvMlC enhances performance relative to single-view approaches by leveraging complementary cross-view information [13], and numerous MvMlC methodologies have emerged in recent years.

However, these MvMlC methods rely on the fundamental premise of complete multi-view multi-label data, which does not hold in empirical reality. In practical deployment, multi-view data acquired from multi-source devices may be incomplete due to factors such as equipment damage; concurrently, manually annotated multi-labels may also be incomplete due to factors such as human omissions. Especially in the field of diagnosing multiple types of diseases, individual doctors are impossible to provide diagnostic results for all diseases based on existing clinical data of patients. Therefore, our research focuses on Partially Multi-View Incomplete Multi-Label Classification (PMvIMlC).

For PMvIMlC, a growing body of research has focused on this issue over the past few years. Tan et al. [14] proposed iMvWL to simultaneously project the partial multi-view features into a shared subspace and learn the structure-aware projection via a weighted linear regression module with the incomplete multi-labels. Li and Chen et al. [15] proposed NAIML, which effectively leverages the available multi-view data consistency and the available multi-label structural information. Although these methods represent significant approaches to PMvIMlC, such traditional machine learning methods often fail to capture underlying discriminative patterns and generalize effectively

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to test data [16]. To address this issue, Wen et al. [16] pioneered deep neural networks (DNNs) for solving the PMvIMlC problem. The method designs an end-to-end architecture and employs autoencoders to derive discriminative latent representations. Subsequently, Liu et al. [13] developed DIC-net, leveraging contrastive learning to strengthen cross-view representation alignment. More recently, Ou et al. [17] proposed an interactive view-label shared Transformer architecture. However, research on PMvIMlC still faces two key challenges: suboptimal coarse-grained multi-view fusion and the neglect of latent information within missing labels.

The first challenge involves suboptimal coarse-grained multi-view fusion. Initially, most research employed multi-view weighted fusion strategies with uniform weights across all views. Subsequent works introduced view-level adaptive fusion, assigning distinct weights to views based on their quality metrics or global attention module [18], [19]. However, such coarse-grained weighting remains sample-agnostic, resulting in suboptimal fusion performance due to the mismatch between view-level weights and individual sample characteristics. In other words, the representation capacity of individual views varies per sample, necessitating sample-specific weight allocation during multi-view fusion based on per-view representation characteristics. The second challenge involves neglecting latent information within missing labels. Most existing approaches handle missing labels by applying mask matrices to exclude unannotated positions, restricting model training to available labels [20], [21]. Considering that the model performance becomes dependent on the amount of available supervision, the majority of existing methods will have performance bottlenecks.

To address the above issues in PMvIMlC task, we propose the **E**vidential **R**eliable **F**usion Network (**ERF** for short). ERF comprises two core components: an uncertainty-guided fusion module via evidence theory and an adaptive negative label pseudo-labeling scheme. The uncertainty-guided fusion module via evidence theory employs evidence theory to quantify the contribution of instance-level features within each view to the classification objective, and allocates adaptive weights based on feature quality metrics to achieve fine-grained fusion. Based on the negative-label predominance assumption in missing labels, the adaptive negative label pseudo-labeling scheme integrates the model's sample similarity perception in label space to implement dynamic screening of reliable negative pseudo-labels. By uncovering latent information in missing labels, this approach incrementally increases supervise information during training, thereby enhancing predictive accuracy.

In summary, our contributions primarily encompass the following points:

- We propose an uncertainty-guided fusion module via evidence theory, effectively addressing suboptimal coarse-grained multi-view fusion. This framework pioneers the application of evidence learning to instance-level multi-view fusion, significantly enhancing the discriminative quality of fused features.
- We propose an adaptive negative label pseudo-labeling scheme that integrates the model's sample similarity perception. With this scheme, the method can reliably recover the negative pseudo-labels, and in turn promotes the performance of the model since the supervisory information is enhanced.

II. Related Works

This section presents the related works regarding PMvIMlL and Uncertainty Quantification.

A. MvMlL

Generally speaking, the core challenges of MvMlL lie in effectively integrating multi-view information and accurately predicting multi-labels. Existing researches have proposed various solutions. For example, to mitigate the constraints of single-view features and the high complexity issues associated with feature concatenation, lrMMC [22] utilizes the low-rank multi-view learning approach to acquire low-dimensional shared representations across diverse views. LSVL [23] aims to ensure structural consistency between the view and label space during feature extraction. Zhang et al. [24] developed the LSA-MML framework to enhance multi-view integration, in which the intrinsic view patterns are aligned in the kernel space to capture the latent semantic structures. To address the issue of ignoring view-specific discriminative information, SIMM [25] extracts view-specific information by imposing orthogonal constraints on the shared subspace. To capture the higher-order relationships and extract the consistency and diversity information among views, CDMM [26] constructs nonlinear kernel mapping classifiers for each view to learn consistent labels and combines the Hilbert-Schmidt independence criterion to explore diversity. Lyu et al. [27] proposed D-VSM, which bypasses shared subspaces and encodes features of each view using deep Graph Convolutional Networks (GCN).

B. PMvL and IMlL

Multi-view multi-label classification boasts broad applicability. However, real-world deployment frequently encounters missing data challenges: partial multi-view missing and partial label missing for any sample in the training and test phases.

For incomplete multi-view data with partially missing views, the prevailing approaches converge upon five distinct methodological branches: matrix decomposition, projection learning, graph learning, kernel learning, and deep learning based methodologies. As a representative matrix-based approach, Yin et al. [28] proposed the IMCRV method, which first reconstructs the absent perspectives of incomplete data and subsequently combines Laplacian regularization to acquire a unified latent space representation. For projection methods, Yin et al. [29] proposed LLE-IMC to acquire a shared representation among multiple views using a weighted projection matrix. For graph-based learning methods, Zhong et al. [30] proposed the SLESC method, where similar graphs are learned for each view. For kernel learning methods, Liu et al. [31] estimated the missing basis matrices using a consistency matrix after clustering, rather than filling the kernel matrices. For deep learning methods, CPSPAN [32] proposes to address the issue of strict alignment

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of representations for the same instance across different views in incomplete multi-view clustering. Recently, Chen et al. [33] proposed a novel method called RIML, which provides an effective approach to handle the issues of missing views and class imbalance by leveraging an inter-class association matrix and category-aware focal loss.

To address the problem of partial label missing, numerous studies have proposed. For example, Wu et al. [34] aimed to restore complete labels for samples by maintaining consistency with existing labels and smoothness in label assignment. C2AE [35] jointly learned feature and category embeddings to explore a potential space to enhance the incomplete multi-label learning performance. Wu et al. [36] proposed a unified hybrid graph based learning model to integrate label dependencies, ensuring training label consistency. Huang et al. [37] presented the LSML method, which initially expands and supplements the label matrix by learning high-order label correlations. Diffent from many methods based on low-order label correlations, CLSML [38] simultaneously completes missing labels and incorporates high-order label correlation information into classifier training.

C. PMvIMlL

For PMvIMlL, existing research can be largely grouped into two principal approaches: traditional non-deep methods (including the matrix decomposition- and projection learning-based methods) and neural network-based methods. Among the traditional methods, the iMVWL method proposed by Tan et al. [14] is one of the representative methods. Its core idea is to simultaneously learn shared subspaces, local label correlations, and subspace predictors, thereby capturing cross-view relationships and weak label information. Building on this, NAIML [15] designs a weighted multi-view weak label regression model, in which the missing label location matrices are introduced to eliminate the influence of missing labels. In addition, it characterizes the global structure of multi-label data as high-rank while depicting the local structure as low-rank to preserve the label structure in the embedding space. Unlike matrix decomposition methods, neural network-based methods leverage deep learning frameworks for end-to-end learning. DD-IMvMLC-net [16] is the first to employ neural networks to address the learning issue of PMvIMlC. DD-IMvMLC-net is based on an encoder-decoder architecture and provides an end-to-end solution for such tasks. To enhance the consistency of multi-view features, Liu et al. [13] proposed DIC-net, which introduces a contrastive learning mechanism to regularize the distance between samples from different views. Unlike the aforementioned methods, MTD decouples the consistency and complementarity features of multi-views while introducing a mask mechanism to bolster the model's capacity for learning. In terms of view-label interaction modeling, Ou et al. [17] proposed a view-category interactive shared Transformer, which uses two Transformer modules to characterize the interaction between views and labels. Considering that most contrastive learning approaches blindly increase the distance between similar negative pairs, Liu et al. [39] further proposed RANK, which introduces a quality-aware subnetwork to assist the multi-view fusion and improves the classification loss by integrating the label relevance. In addition, considering the difference on view information quality, Lu et al. [19] proposed QARF-net, which provides a new approach to dynamic fuse the information of different views according to their view qualities.

D. Uncertainty Quantification and Evidence Learning

For uncertainty quantification, traditional deep neural networks (DNNs) employ parameter point estimation and are thus incapable to produce interpretable uncertainty metrics. Consequently, the research community has developed diverse frameworks to mitigate this limitation, where Bayesian neural networks (BNNs) based method is a prominent approach [40], [41]. BNNs based methods generally transform the parameters of neural networks from point estimates to probability distributions, thereby obtaining the uncertainty of the model. However, these methods face the fact that the posterior distribution is often intractable. Subsequently, Monte Carlo Dropout [42] and Ensemble-based methods [43] were proposed. The former approximates uncertainty by performing dropout operations on a trained neural network with multiple samplings. The latter trains multiple independent models and quantifies the prediction differences among them to calculate the uncertainty. However, both methods suffer from high computational costs. To address the issue, Sensoy et al. proposed an evidential deep learning (EDL) method for measuring uncertainty [44]. This method does not require complex sampling or multi-model training. Specifically, it models the output as a Dirichlet distribution, encodes the "belief mass" and "uncertainty mass" from evidence theory through distribution parameters, and directly models uncertainty using subjective logic theory, simultaneously outputting prediction results and uncertainty measurements in a single forward pass. Owing to the above superior characteristics, this approach has gained significant popularity in recent years for uncertainty measurement [45].

III. The Proposed Method

This section formally defines the PMvIMlC problem and introduces the proposed ERF methodology.

A. Problem Definition and Notation

We use $\{X^{(v)}\}_{v=1}^{m}$ to represent the data with m views, and $X^{(v)} \in R^{n\times d_v}$ is the feature of the v-th view, where n denotes the number of instances, and d_v denotes the feature dimension of the v-th view. We use $Y \in \{0,1\}^{n\times c}$ to represent the corresponding labels, where c denotes the total number of categories in the label space. If $Y_{i,j} = 1$, it indicates that the i-th sample has the j-th label; otherwise, $Y_{i,j} = 0$. For the view missing scenes, we use the missing view guidance matrix $W \in \{0,1\}^{n\times m}$ to mark the location of the missing views. When $W_{i,v} = 0$, it indicates that the v-th view of the i-th sample is missing; otherwise $W_{i,v} = 1$. Similarly, we use a missing label guidance matrix $G \in \{0,1\}^{n\times c}$ to mark the missing labels for all training samples. If $G_{i,j} = 0$, it indicates that the j-th class label

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TABLE I
THE SUMMARY OF NOTATIONS IN THIS PAPER

Symbols	Meanings
n	The number of samples
m	The number of views
c	The number of categories
d_v	The dimension of the i-th view
d	The dimension of the latent subspace
$X^{(v)}$	The original features in the v-th view
Y	The true label matrix
W	Location index matrix of missing views
G	Location index matrix of missing labels
$Encoder^{(v)}$	The encoder in the v-th view
$X'^{(v)}$	Advanced embedded features in the v-th view
Z	The fusion feature
$\overline{Z}$	Feature-enhanced fusion features
$E^{+(v)}$	Evidence matrix for support each category in the v-th view
$E^{-(v)}$	Evidence matrix for oppose each category in the v-th view
$\alpha^{(v)}, \beta^{(v)}$	Shape parameters of the Beta distribution in the v-th view
$\omega^{(v)}$	The subjective opinion in the v-th view
$(B^{+})^{(v)}$	Belief mass matrix in the v-th view
$(B^{-})^{(v)}$	Disbelief mass matrix in the v-th view
$U^{(v)}$	Uncertainty matrix in the v-th view
$\bar{u}^{(v)}$	Instance-level uncertainty for the v-th view
$S^{(v)}$	Beta strength matrix in the v-th view
$\rho^{(v)}$	The multi-view instance-level fusion weight of v-th view
C	Normalized fusion weights matrix
$\Omega^{(v)}$	The ASL error matrix of the v-th view
$Decoder^{(v)}$	The decoder in the v-th view
P	The prediction probability matrix of the fusion feature
Y_l	Observed true labels matrix
$D_{i,j}$	Distance between sample i and j
N_k_i	The set of k nearest neighbor samples of sample i
Υ	the prediction deviation matrix of the known label positions
$\overline{\Upsilon}$	the prediction deviation matrix of the unknown label positions
$f^0_{i,r}$	Frequency of 0 in neighbors of class r of sample i
$f^1_{i,r}$	Frequency of 1 in neighbors of class r of sample i
$A_{i,r}$	Neighbors' dominant label of missing category r of sample i
$\bar{\Upsilon}^0_{i,r}$	Neighbors' dominant negative label prediction deviation corresponding to the missing category r of sample i
$\bar{\Upsilon}^1_{i,r}$	Neighbors' dominant positive label prediction deviation corresponding to the missing category r of sample i
$\bar{\Upsilon}^{neigh}_{i,r}$	neigbor-aware prediction deviation corresponding to the missing category r of sample i
$\bar{\Upsilon}^{self}_{i,r}$	Self-calibration deviation of the missing category r of sample i
$\bar{\Upsilon}^{trust}_{i,r}$	Trustworthy prediction deviation of the missing category r of sample i
τ_i	Sample-level threshold for reliable negative pseudo-labels
U^{pseudo}	Reliable pseudo-label mask matrix

for the i-th sample is missing; otherwise, $G_{i,j} = 1$. In our modal training process, we fill missing views with random noise and fill missing labels with 0. The objective is to design a multi-view multi-label classification network that 1) can be trained by any multi-view data including the above double incomplete multi-view multi-label data and 2) can accurately predict the labels for any incomplete or complete multi-view data. In addition, Table I comprehensively summarizes all mathematical notation employed throughout this paper.

B. Model Architecture of the Proposed Method

As depicted in Fig. 1, we divide the ERF into four modules: feature extraction module, uncertainty-guided fusion module via evidence theory, feature reconstruction module, and adaptive negative label pseudo-labeling.

1) Feature Extraction Module: To extract advanced semantic expressions from specific multi-views and unify different view feature dimensions, we use specific deep autoencoders for each view to obtain their embedded representations, formalized as $\{X'^{(v)} = Encoder^{(v)}(X^{(v)})\}_{v=1}^{m}$. $X'^{(v)} \in R^{n\times d}$ is the embedded feature extracted by the v-th encoder, where d is the unified dimension for multiple views. $Encoder^{(v)}$ denotes the encoder for the v-th view. Architecturally, the encoder consists of four stacked linear layers alternating with Rectified Linear Unit (ReLU) activation functions, $i.e.$, {Linear, ReLU, Linear, ReLU, Linear, ReLU, Linear}. We adopted the similar structure of DD-IMvMLC-net[16], for the v-th view with feature dimension d_v, the output dimensions of the Linear layer are adaptively set to $0.8d_v$, $0.8d_v$, 1500, and d, respectively.

2) Uncertainty-Guided Fusion Module via Evidence Theory: Multi-view feature fusion is a crucial component in PMvMlC, with its quality directly impacting the final classification accuracy. The initial approach employed weighted averaging, emphasizing fairness [16], [25], [46], [47]. However, due to significant quality discrepancies across views, assigning identical fusion weights fails to adequately capture their respective contributions. Consequently, researchers introduced view-level fusion methods, computing weights based on view quality [18], [19]. Although these methods partially enhance the quality of the fused representation, they result in suboptimal coarse-grained fusion because they generally apply static view-level weights uniformly across all samples, lacking instance-specific adaptability. For example, for a low-light sample, the RGB color information is significantly sensitive to the illumination noise, leading to a degradation in its feature quality. Consequently, it is necessary to assign a lower fusion weight to this sample. In contrast, LAB features maintain strong robustness under low-light conditions, exhibiting superior feature quality, and thus should be assigned a higher fusion weight. Therefore, it is important to develop a suitable metric that can evaluate sample quality to guide the instance-level fusion of multi-view data.

Evidential Deep Learning (EDL) establishes a formal metric to quantify the uncertainty inherent in feature quality [44], [48], [49]. From a theoretical perspective, the adoption of evidential uncertainty is motivated not merely by its empirical efficacy, but more fundamentally by the rigorous mathematical framework Evidence Theory provides for multi-source information aggregation. This framework enables the explicit modeling of "vacuity" (i.e., the lack of evidence) [50]. Distinct from conventional fusion approaches, this approach leverages the Dirichlet distribution to map feature quality onto the intensity of belief distributions, thereby yielding an explicit measure of uncertainty. By assigning lower fusion weights to low-quality features (corresponding to a high-uncertainty metric), this approach theoretically guarantees that the fusion process adheres to the consistency logic of evidence combination, allowing the model to intrinsically identify and suppress the detrimental interference of low-quality features. Motivated by this, we propose an uncertainty-guided fusion module via evidence theory to enhance the fused feature quality at the instance level.

Specifically, we employ an Evidential Project module [51] to transform the view-specific features $X'^{(v)}$ (output by the feature encoder) into view-specific evidence. The formula is represented as $\{(E^{+(v)}, E^{-(v)}) = \text{Evidential_Project}(X'^{(v)})\}_{v=1}^{m}$, where $E^{+(v)}, E^{-(v)} \in [0, +\infty)^{n\times c}$. $E^{+(v)}_{i,j}$ denotes the degree

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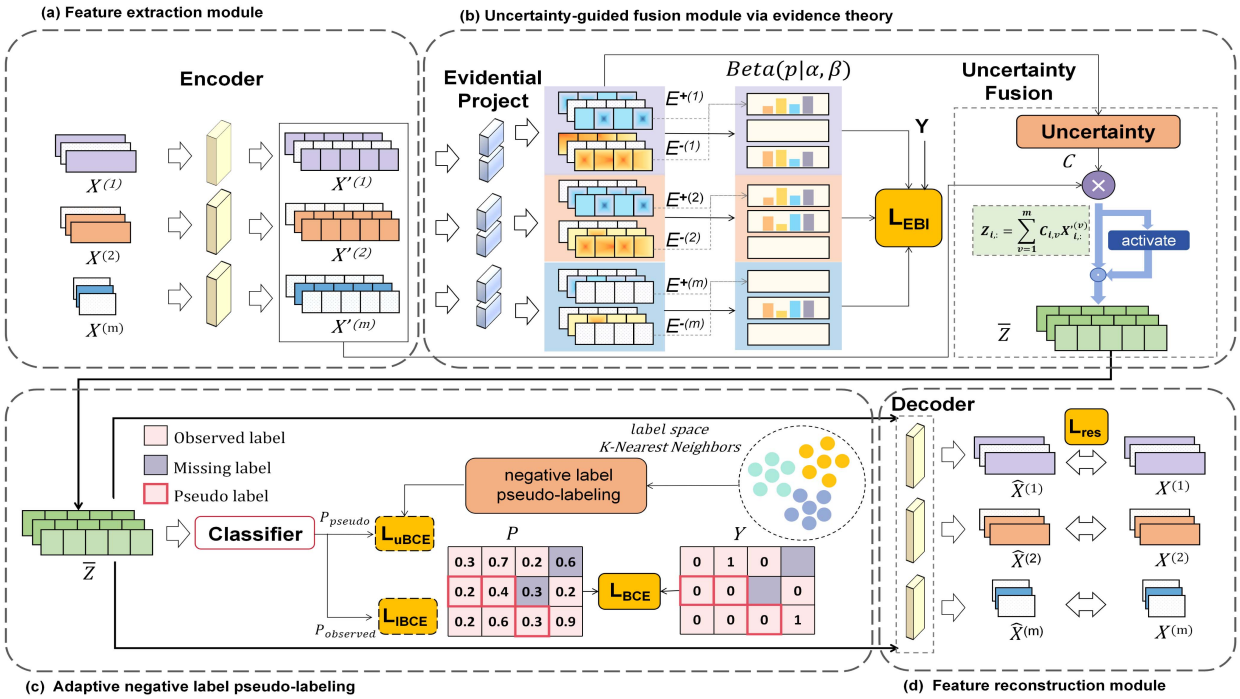


Fig. 1. The framework diagram of ERF, which consists of four modules: (a) Feature extraction module; (b) Uncertainty-guided fusion module via evidence theory; (c) Adaptive negative label pseudo-labeling; (d) Feature reconstruction module. Module (a) extracts the deep semantics of multi-view features and unifies the dimensions of multiple views. Module (b) maps the features of each view to a Beta distribution through an evidence project component, thereby calculating the sample-level fusion weight C to guide sample-level fusion and obtain the fused feature Z. $\overline{Z}$ is the fused feature after feature enhancement via the $sigmoid$ activation function. Module (c) integrates similar samples in the label space, sets reasonable thresholds to filter reliable negative pseudo-labels. $\mathcal{L}_{uBCL}$ and $\mathcal{L}_{lBCL}$ jointly constitute the $\mathcal{L}_{BCE}$ loss function. Module (d) constrains the fused features to contain original necessary information through loss $\mathcal{L}_{res}$.

of support for the proposition that "the i-th sample of the v-th view belongs to the j-th category", and $E_{i,j}^{-(v)}$ represents the strength of opposing the same proposition. Architecturally, the Evidential Project comprises a linear layer followed by a non-negative activation function (e.g., softplus). This design deliberately avoids traditional softmax activation, which is known to induce overconfidence [52], [53], and it enables evidence outputs in $[0, +\infty)$ for subsequent Dirichlet distribution representation.

The Dirichlet distribution is the conjugate prior of the multinomial distribution [54]. Mathematically, the Dirichlet distribution degenerates to a Beta distribution when its dimension is two. Following prior works [48], [55], the relationships between the Beta distribution parameters and evidence are formulated as follows:

$$\alpha^{(v)} = E^{+(v)} + 1, \tag{1}$$

$$\beta^{(v)} = E^{-(v)} + 1. \tag{2}$$

Here, $\alpha^{(v)} \in [1, +\infty)^{n\times c}$ and $\beta^{(v)} \in [1, +\infty)^{n\times c}$ denote the Beta distribution parameters in the v-th view. Among Formulas (1) and (2), the value 1 can be regarded as a non-informative Beta prior. When evidence is missing (e.g. $E_{i,j}^{+(v)} = 0$ or $E_{i,j}^{-(v)} = 0$), it provides unbiased base weights to meet the statistical inference requirements of multi-label learning scenarios [56]. The formal definition of the Beta distribution is provided in Appendix I, available online.

The parameters of the Beta distribution can be mapped to subjective opinions in Subjective Logic. Specifically, a subjective opinion for the v-th view can be represented by $\omega^{(v)} = (B^{+(v)}, B^{-(v)}, U^{(v)})$. $B^{+(v)} \in [0, 1]^{n\times c}$ represents the belief mass of view v, indicating the probability that the sample belongs to the corresponding category. $B^{-(v)} \in [0, 1]^{n\times c}$ denotes the disbelief mass of view v, reflecting the probability that the sample does not belong to the corresponding category. $U^{(v)} \in [0, 1]^{n\times c}$ denotes the uncertainty mass of view v, which captures the probability that a definitive judgment cannot be made due to limitations in feature quality. These components obey the compositional constraint:

$$B_{i,j}^{+(v)} + B_{i,j}^{-(v)} + U_{i,j}^{(v)} = 1. \tag{3}$$

The subjective opinions and evidences have the following relationships:

$$B_{i,j}^{+(v)} = \frac{E_{i,j}^{+(v)}}{S_{i,j}^{(v)}} = \frac{\alpha_{i,j}^{(v)} - 1}{S_{i,j}^{(v)}}, U_{i,j}^{(v)} = \frac{2}{S_{i,j}^{(v)}},$$

$$B_{i,j}^{-(v)} = \frac{E_{i,j}^{-(v)}}{S_{i,j}^{(v)}} = \frac{\beta_{i,j}^{(v)} - 1}{S_{i,j}^{(v)}}, \tag{4}$$

where $S_{i,j}^{(v)} = E_{i,j}^{+(v)} + E_{i,j}^{-(v)} + 2 = \alpha_{i,j}^{(v)} + \beta_{i,j}^{(v)}$ is referred to as Beta strength. Based on Formula (4), we derive the uncertainty mass per class for each view-specific feature representation. Considering the impact of missing views, we employ location

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matrix G to mask out uncertainties at missing positions, obtaining the view-specific instance uncertainty for the v-th view as formalized below:

$$\bar{u}_i^{(v)} = \frac{\sum_{j=1}^{c} U_{i,j}^{(v)} G_{i,j}}{\sum_{j=1}^{c} G_{i,j}}, \tag{5}$$

where $\bar{u}^{(v)} = [\bar{u}_1^{(v)}; \bar{u}_2^{(v)}; \ldots \bar{u}_n^{(v)}] \in [0,1]^{n\times 1}$. $\bar{u}_i^{(v)}$ is the uncertainty of the v-th view of the i-th sample for the task. $\bar{u}^{(v)}$ means instance-level uncertainty for the v-th view. Since the instance-level fusion weights are inversely proportional to their corresponding uncertainties across views, we transform the view-specific fusion weight for the v-th view as:

$$\rho^{(v)} = 1 - \bar{u}^{(v)}, \tag{6}$$

where $\rho^{(v)} \in [0,1]^{n\times 1}$ denotes the instance-level fusion weight for the v-th view. Furthermore, considering that some views may be missing, we normalize the fusion weights as follows:

$$C_{i,:} = \frac{[\rho_i^{(1)}, \ldots, \rho_i^{(m)}] \odot W_{i,:}}{\sum_{v=1}^{m} W_{i,v}}, \tag{7}$$

where $C_{i,:} \in [0,1]^{1\times m}$ represents the fusion weight of the i-th sample. Subsequently, we fuse multi-view features using the normalized fusion weights $C_{i,v}$:

$$Z_{i,j} = \sum_{v=1}^{m} C_{i,v} X'^{(v)}_{i,j}, \tag{8}$$

where $Z \in R^{n\times d}$ is the feature after multi-view fusion. Inspired by [21], we leverage activation functions to boost the fusion features as follows:

$$\overline{Z} = Z \odot \sigma(Z), \tag{9}$$

where σ indicates a sigmoid activation function and where $\overline{Z} \in R^{n\times d}$ denotes the ultimate fusion representation.

To guarantee the accurate quantification of uncertainty, a loss function is needed to regularize the parameters of the Beta distribution. Specifically, in Evidential Deep Learning, the binary cross-entropy (BCE) loss is conventionally applied for this purpose [57]. However, in multi-label scenarios, the number of negative labels in a sample is usually greater than the number of positive labels. This label imbalance may lead to underemphasizing the gradients from positive labels during training [58]. Therefore, inspired by [51], [58], we adopt the following asymmetric Beta distribution loss to guide the optimization of Beta distribution parameters:

$$\mathcal{L}_{\text{EBL}} = \sum_{i=1}^{n} \frac{\sum_{v=1}^{m}\sum_{j=1}^{c} \Omega_{i,j}^{(v)} G_{i,j} W_{i,v}}{\sum_{v=1}^{m}\sum_{j=1}^{c} G_{i,j} W_{i,v}}, \tag{10}$$

where $\Omega_{i,j}^{(v)}$ denotes the asymmetric Beta distribution error for category j of sample i in view v. $\Omega_{i,j}^{(v)}$ is calculated as follows:

$$\Omega_{i,j}^{(v)} = \begin{cases} -\lambda^{+}\left[\psi(\alpha_{i,j}^{(v)} + \beta_{i,j}^{(v)} + \gamma^{+}) - \psi(\alpha_{i,j}^{(v)})\right], & \text{if } Y_{i,j}=1, \\ -\lambda^{-}\left[\psi(\tilde{\alpha}_{i,j}^{(v)} + \beta_{i,j}^{(v)} + \gamma^{-}) - \psi(\beta_{i,j}^{(v)})\right], & \text{if } Y_{i,j}=0, \end{cases} \tag{11}$$

where $\lambda^{+} = \prod_{\gamma=0}^{\gamma^{+}-1} \frac{\beta_{i,j}^{(v)}+\gamma}{\alpha_{i,j}^{(v)}+\beta_{i,j}^{(v)}+\gamma}$ and $\lambda^{-} = \prod_{\gamma=0}^{\gamma^{-}-1} \frac{\tilde{\alpha}_{i,j}^{(v)}+\gamma}{\tilde{\alpha}_{i,j}^{(v)}+\beta_{i,j}^{(v)}+\gamma}$ are weights for adjusting positive and negative class errors. $\tilde{\alpha}_{i,j}^{(v)} = max(\alpha_{i,j}^{(v)} - \frac{h}{1-h}\beta_{i,j}^{(v)}, 0)$ is a shifted random variable [51] for filtering simple negative classes of Beta parameters and h is a constant probability shift parameter. γ^{+} and γ^{-} stand for focusing parameters. In practice, we often set $\gamma^{+} = 0$, $\gamma^{-} > 0$ to focus on positive categories [58]. $\psi(\cdot)$ is the digamma function. The derivation of Formula (11) is provided in Appendix II, available online.

In loss $\mathcal{L}_{\text{EBL}}$, location matrices W and G are introduced to exclude errors at missing positions, aggregating only the errors at observed valid positions to enhance the reliability of the uncertainty quantification.

3) Feature Reconstruction Module: Feature encoders may suffer information loss during the forward propagation process of feature extraction, meaning it's difficult to guarantee that the obtained representation $\overline{Z}$ captures sufficient information from the original data. To address this issue, inspired by [59], we leverage the unique advantages of autoencoders in preserving information by introducing a data reconstruction module $\widehat{X}^{(v)} = Dencoder^{(v)}(\overline{Z})$ for each view and a reconstruction loss for data without missing information, where $\widehat{X}^{(v)} \in R^{n\times d_v}$ are reconstructed features in the v-th view and $Dencoder^{(v)}$ is the decoder of v-th view. For the v-th decoder network, the sizes of the five linear layers are set to d, 1500, $0.8d_v$, $0.8d_v$, and d_v, respectively. The reconstruction loss $\mathcal{L}_{rec}$ for unmissing views are formulated as follows:

$$\mathcal{L}_{rec} = \frac{1}{m}\sum_{v=1}^{m}\Big(\sum_{i=1}^{n}\left\|X_i^{(v)} - \widehat{X_i}^{(v)}\right\|_2^2 W_{i,v}\Big)/nd_v. \tag{12}$$

4) Adaptive Negative Label Pseudo-Labeling: After obtaining reliable multi-view fusion features, we leverage them for multi-label classification. Specifically, the classifier is implemented by a linear layer and a sigmoid activation function, formally expressed as $P = Classifier(\overline{Z})$, where $P \in [0,1]^{n\times c}$ is the prediction probability matrix. Following the approach [16], we optimize the classifier using a weighted binary cross-entropy (BCE) loss, denoted as $\mathcal{L}_{lBCE}$:

$$\begin{aligned}\mathcal{L}_{lBCE} = -\frac{1}{nc}\sum_{i=1}^{n}\sum_{j=1}^{c}\big(&Y_{i,j}\log(P_{i,j}) \\ &+ (1-Y_{i,j})\log(1-P_{i,j})\big)G_{i,j}.\end{aligned} \tag{13}$$

As specified in Formula (13), model training exclusively utilizes available labels by introducing the label-incomplete guidance matrix G.

Indeed, in the model training phase, prior information of known labels can be used to fill the missing labels and the recovered labels can be in turn used to guide the model training so as to further enhance the model performance. This approach seems very reasonable, but there is a key challenge: *how to evaluate the quality of missing label recovery and use reliable recovered labels to improve model performance*? In our paper, different from the conventional approaches that uses all predicted labels

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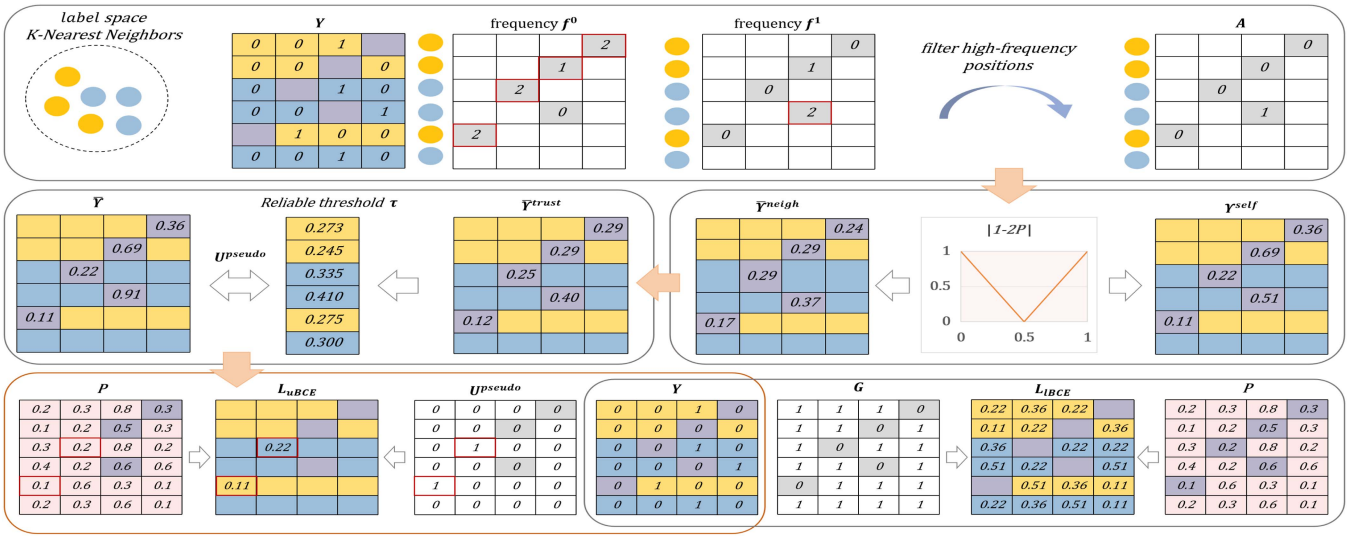
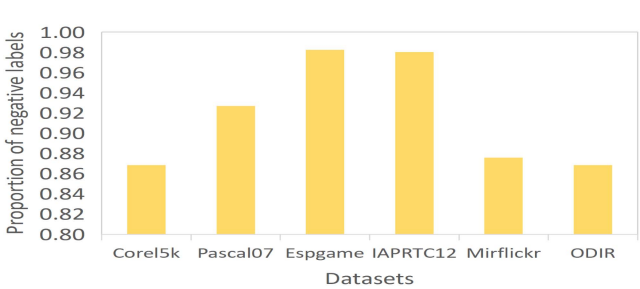
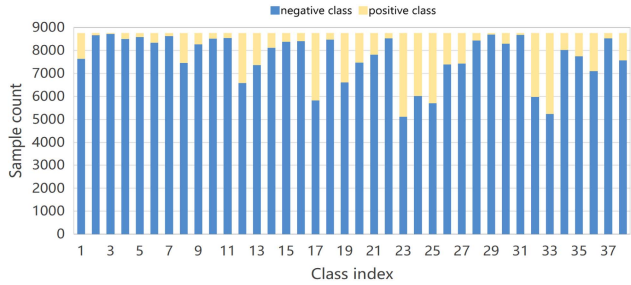


Fig. 2. The workflow of the Adaptive Negative Label Pseudo-labeling strategy. The orange arrows represent our main workflow, while the white arrows indicate the intra-block processes. K-nearest Neighbor (k-NN) is employed to mine similar samples in the label space, and the trustworthy prediction deviation is derived by fusing the sample's own prediction with neighborhood information. Eventually, reliable negative pseudo-labels are filtered via a sample-level dynamic threshold and incorporated into the loss of known labels to augment the supervisory signals.



(a) The proportion of observed negative labels on six datasets.



(b) Number of positive and negative samples per category on the Mirflickr dataset.

Fig. 3. Distribution of positive and negative labels on six popular multi-label datasets.

as supervise guidance, we seek to recover the negative labels and only explore the reliable negative labels to guide the model training. Our motivation stems from statistical analysis of many popular multi-label datasets as shown in Fig. 3. In real-world multi-label data, the proportion of negative labels is usually significantly higher than that of positive labels. In other words, the number of label tags that a sample possesses at the same time often accounts for only a small portion of the total number of label tags in the dataset. Therefore, in multi-label scenarios with missing labels, the number of missing negative labels is often significantly greater than the number of missing positive labels. Obviously, due to the larger proportion of negative labels, recovering them and finding the reliable recovered labels are often easier than those of for the positive labels. Inspired by this motivation, we design an adaptive negative pseudo-labeling strategy as shown in Fig. 2 that incorporates reliable negative pseudo-label supervisory information during training. Specifically, we first compute prediction losses for observed labels and missing labels (filled by 0) separately using the Binary Cross-Entropy (BCE) loss, as formalized below:

$$\Upsilon = (Y \odot \log(P) + (1 - Y) \odot \log(1 - P)) \odot G, \tag{14}$$

$$\overline{\Upsilon} = (Y \odot \log(P) + (1 - Y) \odot \log(1 - P)) \odot (1 - G), \tag{15}$$

where $\Upsilon, \overline{\Upsilon} \in R^{n \times c}$. Υ and $\overline{\Upsilon}$ indicate the prediction deviation between the prediction labels and the ground labels at the known and unknown label positions, respectively.

Furthermore, we propose a simple sample-level threshold based method to select the trustworthy recovered labels. According to the premise that similar samples tend to exhibit analogous label distributions and exhibit similar prediction performance, in which the latter can be qualified by prediction deviation between the prediction labels and the ground labels [60]. Firstly, we find k Nearest Neighbors (k-NN) samples in the label space Y for each sample, as shown in the following formulas:

$$D_{i,j} = \|Y_{i,:} - Y_{j,:}\|_2^2, \tag{16}$$

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$$N_k_i = \{j \in J \mid j \neq i \wedge D_{i,j} \leq top(i,k)\}, \tag{17}$$

where $D \in \mathbb{R}^{n\times n}$ and $D_{i,j}$ represents the distance between the i-th sample and the j-th sample in the label space. $J = \{1, 2, \ldots, n\}$, and N_k_i denotes the set of k nearest neighbor samples for sample i. $top(i,k)$ is the top k value of $D_{i,:}$.

Given that the labels observed in the neighbor samples are not completely identical, we only consider the prediction deviation of the model on the dominant labels (0 or 1) in the k-NN set. Specifically, we calculate the neighbors' dominant labels for missing positions in the anchor sample i as follows:

$$f_{i,r}^0 = \left(\sum_{j\in N_k_i}(1-Y_{j,r})G_{j,r}\right)(1-G_{i,r}), \tag{18}$$

$$f_{i,r}^1 = \left(\sum_{j\in N_k_i} Y_{j,r}G_{j,r}\right)(1-G_{i,r}), \tag{19}$$

$$A_{i,r} = \begin{cases} 1, & \text{if } \left(f_{i,r}^1 > f_{i,r}^0\right), \\ 0, & otherwise. \end{cases} \tag{20}$$

Among them, $f_{i,r}^0$ and $f_{i,r}^1$ represent the negative and positive label frequency of the neighbors corresponding to missing category r of sample i, respectively. $A_{i,r}$ represents the neighbors' dominant label of missing category r of sample i. Then, we calculate the prediction deviation of the model on the dominant label in the neighbors as the perception of the model's prediction deviation for missing sample categories, as follows:

$$\bar{\Upsilon}_{i,r}^0 = \frac{\sum_{j\in N_k_i}\Upsilon_{j,r}(1-Y_{j,r})(1-A_{i,r})(1-G_{i,r})}{\sum_{j\in N_k_i}(1-Y_{j,r})(1-A_{i,r})(1-G_{i,r})}, \tag{21}$$

$$\bar{\Upsilon}_{i,r}^1 = \frac{\sum_{j\in N_k_i}\Upsilon_{j,r}Y_{j,r}A_{i,r}}{\sum_{j\in N_k_i}Y_{j,r}A_{i,r}}, \tag{22}$$

where $\bar{\Upsilon}_{i,r}^0$ and $\bar{\Upsilon}_{i,r}^1$ are the neighbors' prediction deviation of neighbors' dominated negative labels and positive labels corresponding to the missing category r of sample i, respectively. Therefore, the neighbor-aware prediction deviation $\bar{\Upsilon}^{neigh}$ corresponding to the missing label positions can be calculated as follows:

$$\bar{\Upsilon}^{neigh} = \bar{\Upsilon}^0 + \bar{\Upsilon}^1. \tag{23}$$

However, neighbor-aware prediction deviation exhibits a heterogeneous distribution owing to the intrinsic learning complexity of samples. Therefore, to obtain reliable negative pseudo-labels, the neighbor-aware prediction deviation for missing positions should be adaptively calibrated by incorporating the model's predictions on these positions. Consequently, we design a self-calibration deviation Υ^{self}, formulated as follows:

$$\begin{aligned}\Upsilon_{i,r}^{self} = &\left(A_{i,r}\log\left(P_{i,r}\right)\right. \\ &\left.+\left(1-A_{i,r}\right)\left(1-G_{i,r}\right)\log\left(1-P_{i,r}\right)\right)\left(1-G_{i,r}\right),\end{aligned} \tag{24}$$

where $\Upsilon_{i,r}^{self}$ denotes the self-calibration deviation of the missing category r of the i-sample.

Subsequently, we fuse the neighbor-aware prediction deviation and self-calibration deviation based on the prediction probability P. We use $|1-2P|$ as the weight to balance the self-calibration deviation and neighbor prediction deviation. When $P_{i,j} = 0.5$, indicating that the prediction at the current position is untrustworthy, the neighbor prediction deviation is directly estimated as the trust prediction deviation of the missing position; when $P_{i,j}$ is close to 0 or 1, the neighbor prediction deviation is given less consideration. The formalization is defined below:

$$\bar{\Upsilon}_{i,r}^{trust} = |1-2P_{i,r}|\,\Upsilon_{i,r}^{self} + (1-|1-2P_{i,r}|)\,\bar{\Upsilon}_{i,r}^{neigh}, \tag{25}$$

where $\bar{\Upsilon}_{i,r}^{trust}$ denotes the trustworthy prediction deviation of the missing position r of the i-th sample.

Thereafter, we compute the sample-level threshold to determine the reliability of the negative pseudo-labels. The formula is as follows:

$$\tau_i = \frac{1}{c}\sum_{r=1}^{c}(\bar{\Upsilon}_{i,r}^{trust}(1-G_{i,r}) + \Upsilon_{i,r}G_{i,r}), \tag{26}$$

where τ_i is the reliable threshold for the i-th sample. For negative pseudo-labels, if the prediction deviation at the missing position is less than the threshold, it is considered trustworthy. The formulas are as follows:

$$U_{i,r} = \begin{cases} 1, & \text{if } \left(\overline{\Upsilon}_{i,r} < \tau_i\right), \\ 0, & otherwise. \end{cases} \tag{27}$$

$$U_{i,r}^{\text{pseudo}} = U_{i,r}(1-G_{i,r}), \tag{28}$$

where U is the location matrix that satisfies the reliability threshold, and U^{pseudo} is a mask matrix for pseudo label. $U_{i,r}^{\text{pseudo}}$ mask reliable negative pseudo-labels of the category r of sample i. At this point, the classification loss consists of two parts: the classification loss of observed labels $\mathcal{L}_{lBCE}$ (13) and the loss of reliable negative pseudo-labels $\mathcal{L}_{uBCE}$, as shown in the following formula:

$$\begin{aligned}\mathcal{L}_{uBCE} = &-\frac{1}{nc}\sum_{i=1}^{n}\sum_{j=1}^{c}\left(Y_{i,j}\log\left(P_{i,j}\right)\right. \\ &\left.+\left(1-Y_{i,j}\right)\log\left(1-P_{i,j}\right)\right)U_{i,j}^{\text{pseudo}}.\end{aligned} \tag{29}$$

The total classification loss is $\mathcal{L}_{BCE} = \mathcal{L}_{lBCE} + \mathcal{L}_{uBCE}$.

During training, as the model remains unstable in the initial stages, a warm-up strategy is employed, with the pseudo-label strategy activated only after training for a predefined number of epochs ε.

5) Total Loss Function: The total loss includes three parts: classification loss, evidence loss, and reconstruction loss, with the formula given as follows.

$$\mathcal{L}_{all} = \mathcal{L}_{BCL} + \gamma\mathcal{L}_{rec} + \theta\mathcal{L}_{EBL}, \tag{30}$$

where γ and θ are penalty coefficients.

Our model training process is shown in the pseudocode of Algorithm 1.

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Algorithm 1: Training Procedure of the ERF Model

Input: Partial multi-view data $\{X^{(v)}\}_{v=1}^{m}$ and the associated incomplete label matrix Y, missing-view indication matrix W, missing-label indication matrix G.
Output: Prediction P
Initialize model parameters and hyperparameters (learning rate lr, batch size B, maximum number of iterations T, γ, θ, h, γ^{+}, γ^{-}, k, and warm-up ε).
for t=1 to T **do**
  Extracting multi-view advanced embedded features $X^{'(v)}$ through $Encoder^{(v)}$;
  Implementing uncertainty-guided fusion module to fusion multi-view embedded features by (8);
  Calculating evidence loss $\mathcal{L}_{EBL}$ by (10);
  Reconstructing data by $Decoder^{(v)}$;
Calculating the reconstruction loss $\mathcal{L}_{rec}$ by (12);
  if $t > \varepsilon$:
  Start pseudo label mechanism and calculate $\mathcal{L}_{uBCL}$ by (29);
  Calculating classification loss $\mathcal{L}_{lBCL}$ by (13);
  Calculating the total loss $\mathcal{L}_{all}$ via (30);
  Updating the network parameters.
end for
return model

IV. Experiments

A. Experiment Configuration

1) Datasets: To comprehensively evaluate the performance of our ERF method, we conducted experiments on several benchmark datasets, which are widely used in the field of partial multi-view incomplete multi-label learning. They are Corel5k [61], Iaprtc12 [62], Espgame [63], Mirflicker [64], and Pascal07 [65]. Additionally, we have incorporated the ODIR-5k [66] dataset for fundus disease diagnosis to further validate the effectiveness of the ERF model. The details of these six datasets are provided in Appendix III, available online.

2) Compared Methods: We selected nine commonly used models as comparative algorithms and comprehensively analyzed the performance of our ERF model. The selected methods include **C2AE** [35], **LVSL** [23], **iMvWL** [14], **NAIML** [15], **LMVCAT** [18], **DIC-net** [13], **MTD** [21], **DD-IMvMLC-net** [16], and **RANK** [39]. Section II contains detailed elaborations on these approaches. More details of the compared methods are listed in Appendix IV, available online.

3) Evaluation Metrics: In alignment with previous research [13], [14], [15], [16], [18], [39], we adopt three standard evaluation metrics to evaluate model effectiveness across the six datasets: Average Precision (AP), Hamming Loss (HL), and One-Error (OE). For intuitive comparison, we transform HL and OE into 1-HL and 1-OE, respectively, ensuring that higher values uniformly indicate superior model performance.

4) Implementation Details: We implement our ERF model in Python using PyTorch, training it with an SGD optimizer configured with 0.9 momentum. The model is implemented on an Ubuntu environment using an RTX 4080 GPU and an i7-13700F CPU. All trainable parameters of the model (full connection layer and BatchNormld layer) adopt PyTorch built-in default initialization rules. Please refer to Appendix V for more details.

B. Experimental Results and Analysis

To validate the efficacy of our ERF, we conducted validation experiments under three settings: 50% partial views with 50% incomplete labels (Table II); single-missing scenarios (Table III); and the complete multi-view multi-label scenario (Table IV). The experimental results of some comparison methods are derived from [18], [21], [39].

- Table II compares the key indicators for nine models in the PMvIMlC scenario, and the results demonstrate that the proposed method achieves superior performance across all indicators. For example, on the Corel5k dataset, the AP indicator improves by more than 2% compared to the second-best RANK. These experimental outcomes confirm the proposed methodology's advantage over comparative methods in PMvIMlC scenarios.
- Table III confirms our model's efficacy in scenarios with only a single type of missing data. Specifically, on Corel5k with only 50% missing labels, our ERF outperforms the second-best RANK by 4.6% in AP. Similarly, in scenarios with partial multi-view but complete labels, ERF achieves a nearly 3% improvement over RANK, validating its effectiveness in semi-missing conditions.
- As shown in Table IV, even in scenarios without missing values, our ERF framework still demonstrates significant advantages in MvMlC. Specifically, on the Corel5k dataset, the AP metric of ERF is improved by more than 3% compared with the second-best RANK model.
- Fig. 4 provides a horizontal comparison of five baseline methods across four missing-data settings: dual-missing (PMvIMlC), complete multi-view incomplete multi-label (MvIMlC), partial multi-view with complete label (PMvMlC), and dual-complete (MvMlC). The results show that the complete-view setting MvIMlC generally yields better performance than the complete-label setting PMvMlC. Moreover, our ERF achieves the best results under all missingness settings.

C. Ablation Study

In this study, we conduct a comprehensive ablation analysis to validate the effectiveness of each proposed module. Additionally, we replace the average-weighted fusion in the Backbone with the view-level fusion method from [19], termed QARF_fusion, to validate the superiority of the proposed uncertainty-guided fusion module.

The ablation study results are shown in Table V. First, replacing the uncertainty-guided fusion module via evidence theory with a simple weighted-average fusion (Backbone) leads to a drop of 1.2 percentage points in AP on Pascal07. Compared to the view-level fusion (QARF_fusion), our method also exhibits a similar trend of performance improvement. This improvement

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TABLE II
THE EXPERIMENTAL RESULTS OF TEN DIFFERENT APPROACHES ACROSS SIX DATASETS UNDER SETTINGS OF 50% ABSENCE OF VIEWS, 50% ABSENCE OF LABELS, AND 70% TRAINING SAMPLES. THE SYMBOL ↑ DENOTES THAT GREATER VALUES CORRESPOND TO BETTER MODEL EFFICACY.

DATA	METRIC	C2AE	LVSL	iMvWL	NAIML	DD-IMvMLC-net	LMVCAT	DIC-net	MTD	RANK	ERF
Corel5k	AP↑	0.227±0.008	0.342±0.004	0.283±0.008	0.309±0.004	0.353±0.006	0.382±0.004	0.381±0.004	0.415±0.008	0.427±0.006	**0.448±0.006**
	1-HL↑	0.980±0.002	0.987±0.000	0.978±0.000	0.987±0.000	0.987±0.000	0.986±0.000	0.988±0.000	0.988±0.000	0.988±0.000	**0.988±0.000**
	1-OE↑	0.246±0.016	0.391±0.009	0.311±0.015	0.350±0.009	0.422±0.015	0.453±0.006	0.468±0.007	0.491±0.012	0.499±0.012	**0.535±0.007**
Pascal07	AP↑	0.485±0.008	0.504±0.005	0.437±0.018	0.488±0.003	0.532±0.002	0.519±0.005	0.505±0.012	0.551±0.004	0.555±0.010	**0.565±0.007**
	1-HL↑	0.908±0.002	0.930±0.000	0.882±0.004	0.928±0.001	0.931±0.001	0.924±0.003	0.929±0.001	0.932±0.001	0.933±0.001	**0.934±0.001**
	1-OE↑	0.438±0.008	0.419±0.008	0.362±0.023	0.421±0.006	0.456±0.011	0.422±0.006	0.427±0.015	0.459±0.007	0.463±0.016	**0.483±0.009**
Espgame	AP↑	0.202±0.006	0.285±0.003	0.244±0.005	0.246±0.002	0.287±0.002	0.294±0.004	0.297±0.002	0.306±0.003	0.314±0.004	**0.317±0.003**
	1-HL↑	0.971±0.002	0.983±0.000	0.972±0.000	0.983±0.000	0.983±0.000	0.983±0.000	0.983±0.000	0.983±0.000	0.983±0.000	**0.983±0.000**
	1-OE↑	0.262±0.018	0.389±0.004	0.343±0.013	0.339±0.003	0.435±0.009	0.434±0.009	0.439±0.007	0.447±0.009	0.462±0.015	**0.465 ±0.008**
IAPRTC12	AP↑	0.224±0.007	0.304±0.004	0.237±0.003	0.261±0.001	0.308±0.001	0.317±0.003	0.323±0.001	0.332±0.003	0.346±0.004	**0.348±0.003**
	1-HL↑	0.965±0.002	0.981±0.000	0.969±0.000	0.980±0.000	0.980±0.000	0.980±0.000	0.981±0.000	0.981±0.000	0.981±0.000	**0.981±0.000**
	1-OE↑	0.300±0.031	0.429±0.009	0.352±0.008	0.390±0.005	0.431±0.006	0.443±0.005	0.468±0.002	0.467±0.005	0.485±0.006	**0.488±0.008**
Mirflickr	AP↑	0.505±0.008	0.553±0.002	0.490±0.012	0.551±0.002	0.602±0.002	0.594±0.005	0.589±0.005	0.607±0.004	0.602±0.007	**0.610±0.003**
	1-HL↑	0.853±0.004	0.885±0.001	0.839±0.002	0.882±0.001	0.888±0.000	0.882±0.002	0.888±0.002	0.891±0.001	0.890±0.001	**0.892±0.001**
	1-OE↑	0.505±0.020	0.607±0.004	0.511±0.022	0.585±0.003	0.647±0.007	0.643±0.008	0.637±0.007	0.655±0.004	0.653±0.009	**0.664±0.006**
ODIR-5k	AP↑	0.657±0.011	0.667±0.004	0.614±0.007	0.594±0.007	0.671±0.007	0.663±0.013	0.661±0.010	0.680±0.006	0.677±0.007	**0.682±0.006**
	1-HL↑	0.823±0.042	0.873±0.001	0.781±0.005	0.813±0.008	0.875±0.003	0.865±0.005	0.874±0.002	0.876±0.003	0.876±0.002	**0.877±0.003**
	1-OE↑	0.474±0.015	0.454±0.008	0.395±0.010	0.389±0.007	0.477±0.013	0.478±0.017	0.466±0.014	0.490±0.010	0.487±0.010	**0.495±0.010**

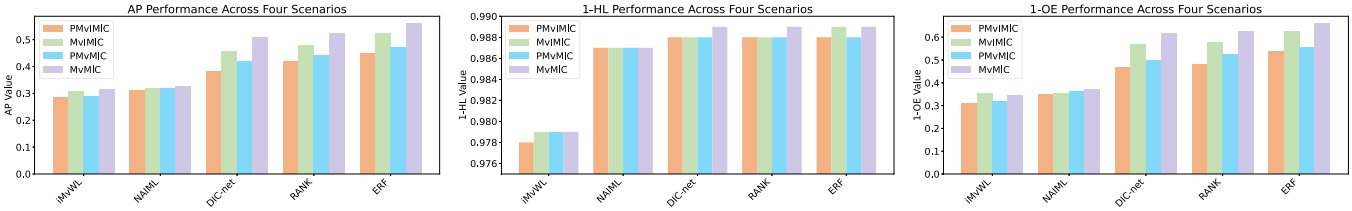
TABLE III
THE OUTCOMES OF THE FIVE METHODOLOGIES ACROSS SIX DISTINCT DATASETS UNDER CONDITIONS OF FULL VIEWS WITH 50% LABEL ABSENCE AND COMPLETE LABELS WITH 50% VIEW ABSENCE

DATA	METRIC	complete views and 50% missing labels					50% missing views and complete labels				
		iMvWL	NAIML	DIC-net	RANK	ERF	iMvWL	NAIML	DIC-net	RANK	ERF
Corel5k	AP↑	0.311±0.004	0.319±0.004	0.457±0.011	0.478±0.011	**0.524±0.004**	0.289±0.009	0.317±0.003	0.419±0.006	0.442±0.009	**0.471±0.005**
	1-HL↑	0.979±0.000	0.987±0.000	0.988±0.000	0.988±0.000	**0.989±0.000**	0.979±0.000	0.987±0.000	0.988±0.000	0.988±0.000	**0.988±0.000**
	1-OE↑	0.352±0.011	0.363±0.010	0.568±0.010	0.579±0.016	**0.624±0.010**	0.320±0.015	0.360±0.008	0.499±0.005	0.525±0.014	**0.557±0.009**
Pascal07	AP↑	0.467±0.005	0.493±0.003	0.562±0.007	0.634±0.004	**0.655±0.009**	0.439±0.013	0.492±0.003	0.525±0.003	0.574±0.009	**0.585±0.008**
	1-HL↑	0.888±0.001	0.929±0.001	0.934±0.000	0.937±0.001	**0.939±0.001**	0.882±0.004	0.929±0.001	0.932±0.002	0.934±0.001	**0.936±0.001**
	1-OE↑	0.410±0.007	0.424±0.007	0.472±0.001	0.544±0.008	**0.575±0.011**	0.358±0.023	0.424±0.006	0.440±0.001	0.482±0.012	**0.506±0.011**
Espgame	AP↑	0.260±0.004	0.249±0.002	0.348±0.000	0.358±0.006	**0.362±0.002**	0.246±0.005	0.249±0.002	0.315±0.003	0.327±0.005	**0.337±0.003**
	1-HL↑	0.972±0.000	0.983±0.000	0.983±0.000	0.983±0.000	**0.983±0.000**	0.972±0.000	0.983±0.000	0.983±0.000	0.983±0.000	**0.983±0.000**
	1-OE↑	0.389±0.011	0.343±0.004	0.509±0.005	0.522±0.008	**0.526±0.005**	0.346±0.013	0.342±0.003	0.464±0.006	0.478±0.010	**0.483±0.007**
IAPRTC12	AP↑	0.249±0.004	0.265±0.001	0.368±0.004	0.395±0.007	**0.397±0.002**	0.236±0.005	0.265±0.001	0.347±0.002	0.355±0.004	**0.368±0.004**
	1-HL↑	0.969±0.000	0.981±0.000	0.981±0.000	0.981±0.000	**0.982±0.000**	0.968±0.000	0.981±0.000	0.981±0.000	0.981±0.000	**0.981±0.000**
	1-OE↑	0.378±0.010	0.394±0.004	0.525±0.001	0.556±0.012	**0.559±0.004**	0.351±0.009	0.393±0.005	0.491±0.014	0.494±0.008	**0.511±0.007**
Mirflickr	AP↑	0.500±0.019	0.554±0.001	0.648±0.001	**0.666±0.004**	0.662±0.003	0.497±0.009	0.553±0.002	0.588±0.003	0.613±0.006	**0.615±0.004**
	1-HL↑	0.842±0.004	0.882±0.001	0.900±0.000	**0.903±0.001**	0.901±0.001	0.840±0.003	0.882±0.001	0.890±0.000	0.893±0.001	**0.893±0.001**
	1-OE↑	0.523±0.018	0.588±0.003	0.719±0.001	0.730±0.006	**0.732±0.005**	0.513±0.019	0.588±0.003	0.638±0.003	0.664±0.008	**0.674 ±0.005**
ODIR-5k	AP↑	0.643±0.005	0.614±0.012	0.684±0.008	0.713±0.008	**0.724±0.005**	0.617±0.011	0.673±0.004	0.667±0.008	0.683±0.005	**0.687±0.010**
	1-HL↑	0.789±0.002	0.823±0.011	0.879±0.008	0.884±0.002	**0.888±0.002**	0.782±0.002	0.874±0.001	0.875±0.002	**0.879±0.002**	0.878±0.003
	1-OE↑	0.430±0.008	0.409±0.020	0.505±0.015	0.539±0.014	**0.557±0.009**	0.398±0.018	0.465±0.006	0.475±0.012	0.495±0.009	**0.498±0.017**

TABLE IV
THE EXPERIMENTAL RESULTS OF TEN DIFFERENT APPROACHES ACROSS SIX DATASETS UNDER COMPLETE VIEWS AND LABELS, WITH 70% TRAINING SAMPLES. THE SYMBOL ↑ DENOTES THAT GREATER VALUES CORRESPOND TO BETTER MODEL EFFICACY.

DATA	METRIC	C2AE	LVSL	iMvWL	NAIML	DD-IMvMLC-net	LMVCAT	DIC-net	MTD	RANK	ERF
Corel5k	AP↑	0.353±0.033	0.482±0.003	0.313±0.005	0.327±0.004	0.506±0.010	0.521±0.005	0.509±0.002	0.523±0.003	0.524±0.008	**0.559±0.004**
	1-HL↑	0.969±0.002	0.988±0.000	0.979±0.000	0.987±0.000	0.989±0.000	0.988±0.000	0.989±0.000	0.989±0.000	0.989±0.000	**0.989±0.000**
	1-OE↑	0.434±0.042	0.554±0.003	0.356±0.010	0.371±0.009	0.606±0.018	0.618±0.008	0.615±0.007	0.629±0.007	0.627±0.013	**0.659±0.005**
Pascal07	AP↑	0.584±0.007	0.588±0.006	0.468±0.004	0.496±0.003	0.668±0.004	0.629±0.007	0.608±0.003	0.648±0.006	0.656±0.007	**0.682±0.005**
	1-HL↑	0.921±0.002	0.934±0.001	0.888±0.001	0.929±0.001	0.941±0.001	0.927±0.005	0.937±0.001	0.940±0.001	0.940±0.001	**0.942±0.001**
	1-OE↑	0.510±0.009	0.483±0.008	0.412±0.005	0.426±0.006	0.590±0.007	0.534±0.011	0.526±0.005	0.562±0.008	0.574±0.013	**0.611±0.008**
Espgame	AP↑	0.269±0.003	0.363±0.002	0.260±0.004	0.251±0.002	0.343±0.003	0.385±0.003	0.391±0.001	**0.397±0.002**	0.388±0.003	0.386±0.003
	1-HL↑	0.958±0.002	0.983±0.000	0.972±0.000	0.983±0.000	0.983±0.000	0.982±0.000	0.984±0.000	**0.984±0.000**	0.984±0.000	0.982±0.000
	1-OE↑	0.354±0.006	0.505±0.004	0.390±0.011	0.345±0.003	0.511±0.006	0.540±0.005	0.549±0.002	0.554±0.005	0.546±0.006	**0.598±0.003**
IAPRTC12	AP↑	0.316±0.004	0.383±0.003	0.250±0.004	0.267±0.001	0.424±0.004	0.436±0.004	0.418±0.006	**0.443±0.003**	0.428±0.005	0.438±0.003
	1-HL↑	0.953±0.001	0.982±0.000	0.969±0.000	0.980±0.000	0.982±0.000	0.981±0.000	0.982±0.000	0.982±0.000	0.982±0.000	**0.982±0.000**
	1-OE↑	0.441±0.010	0.505±0.006	0.380±0.010	0.396±0.004	0.576±0.009	0.581±0.007	0.581±0.001	**0.601±0.005**	0.581±0.008	0.598±0.003
Mirflickr	AP↑	0.593±0.004	0.638±0.002	0.493±0.022	0.555±0.002	0.676±0.004	**0.684±0.006**	0.659±0.004	0.681±0.004	0.680±0.005	0.670±0.003
	1-HL↑	0.869±0.002	0.895±0.001	0.840±0.004	0.882±0.001	0.904±0.001	0.897±0.005	0.902±0.001	**0.906±0.001**	0.905±0.001	0.903±0.001
	1-OE↑	0.641±0.020	0.714±0.004	0.518±0.040	0.590±0.003	0.745±0.006	0.747±0.007	0.728±0.008	**0.749±0.005**	0.747±0.007	0.744±0.006
ODIR-5k	AP↑	0.705±0.005	0.709±0.005	0.634±0.015	0.676±0.005	0.728±0.005	0.721±0.011	0.700±0.010	0.727±0.005	0.729±0.006	**0.732±0.006**
	1-HL↑	0.877±0.003	0.883±0.002	0.789±0.002	0.874±0.001	0.890±0.002	0.881±0.005	0.885±0.003	0.891±0.002	0.890±0.002	**0.892±0.002**
	1-OE↑	0.554±0.005	0.512±0.010	0.415±0.023	0.468±0.006	0.559±0.008	0.555±0.016	0.531±0.018	0.562±0.008	0.562±0.011	**0.570±0.011**

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(a) AP (b) 1-HL (c) 1-OE

Fig. 4. Comparison between the ERF method and four classical multi-view multi-label classification methods across three key metrics under four scenarios on the Corel5k dataset. Specifically, 'PMvIMlC' corresponds to the scenario with 50% partial multi-view data and 50% incomplete multi-label data; 'MvIMlC' refers to the scenario with complete multi-view data and 50% incomplete multi-label data; 'PMvMlC' denotes the scenario with 50% partial multi-view data and complete multi-label data; and 'MvMlC' represents the scenario with complete multi-view data and complete multi-label data.

TABLE V
ABLATION EXPERIMENTAL RESULTS OF THE ERF MODEL ON THE COREL5K AND PASCAL07 DATASETS

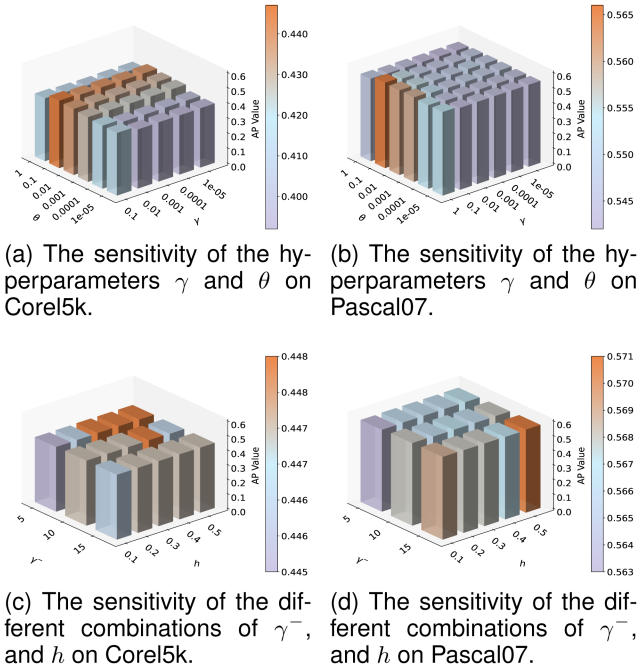
Backbone	evi_fusion	pso_label	Corel5k			Pascal07		
			AP↑	1-HL↑	1-OE↑	AP↑	1-HL↑	1-OE↑
✓			0.432	0.988	0.513	0.551	0.932	0.463
✓	✓		0.442	0.988	0.528	0.563	0.935	0.483
✓		✓	0.433	0.988	0.515	0.554	0.932	0.467
✓	✓	✓	0.448	0.988	0.531	0.565	0.934	0.483
QARF_fusion			0.435	0.988	0.517	0.555	0.933	0.466

is attributed to the instance-level dynamic fusion mechanism grounded in evidence theory, which adaptively captures inherent feature uncertainty to suppress the deleterious impact of low-quality features. Secondly, introducing an adaptive negative label pseudo-labeling strategy to the uncertainty-guided fusion module improves the AP on the Corel5k dataset by 0.5 percentage points, which also validate the effectiveness of the pseudo-labeling strategy.

D. Sensitivity Studies of Hyperparameters

Given that the penalty coefficients γ and θ in the loss function directly influence the reconstruction loss and evidence loss, we conducted hyperparameter sensitivity analyses to quantify their impact. In addition, the ASL loss incorporates three hyperparameters: γ^+, γ^-, and h. To streamline hyperparameter analysis, [58] introduces a simple and intuitive method. By setting $\gamma^+ = 0$, the adjustment is achieved with only a single interpretable control parameter. Consequently, we designed supplementary experiments to investigate how the remaining parameters γ^- and h modulate this balancing capability. Finally, we also conducted experiments on the hyperparameter k of k-nearest neighbors, and the experimental results and analysis are shown in Appendix VI, available online. All hyperparameter evaluations were conducted on the Corel5k and Pascal07 datasets under the scenario of 50% partial view missing and 50% label missing, with the results presented as follows:

- Fig. 5 (a-b) display hyperparameters γ and θ sensitivity across ranges $\gamma \in [10^{-5}, 10^{-1}], \theta \in [10^{-5}, 1]$ on Corel5k; $\gamma, \theta \in [10^{-5}, 1]$ on Pascal07. Results indicate peak performance at $\gamma = 0.1$, $\theta = 0.1$ on Corel5k, and at $\gamma = 1, \theta = 0.1$ on Pascal07.



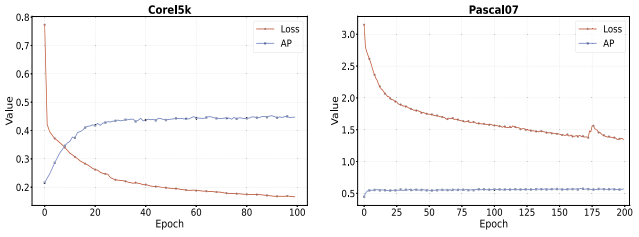
(a) The sensitivity of the hyperparameters γ and θ on Corel5k.

(b) The sensitivity of the hyperparameters γ and θ on Pascal07.

(c) The sensitivity of the different combinations of γ^-, and h on Corel5k.

(d) The sensitivity of the different combinations of γ^-, and h on Pascal07.

Fig. 5. A systematic evaluation of hyperparameter sensitivity was conducted for the ERF model on the Corel5k and Pascal07 benchmarks.



(a) Convergence of the ERF model on Corel5k.

(b) Convergence of the ERF model on Pascal07.

Fig. 6. Convergence curve of our model on the Corel5k and Pascal07 datasets.

- Fig. 5 (c-d) show the hyperparameter experimental results for parameters γ^- and h, with their value ranges being $\gamma^- \in [5, 15]$ and $h \in [0.1, 0.5]$, respectively. The experimental results indicate that on the Corel5k dataset, when

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TABLE VI
THE EXPENSE OF TEMPORAL RESOURCES DURING THE TRAINING AND INFERENCE STAGES ON THE COREL5K DATASET, UTILIZING 70% OF THE SAMPLES FOR TRAINING. (UNIT: SECOND)

Stages	C2AE	LVSL	iMVWL	NAIML	DD-IMvMLC-net	LMVCAT	DIC-net	MTD	RANK	ERF
Training	170.24	63.73	165.82	143.63	141.85	238.00	313.89	137.00	450.51	213.97
Inference	0.04	0.64	0.02	0.01	0.04	0.04	0.05	0.69	0.35	0.35

the value of γ^- is close to 5 and the value of h is in the interval of [0.3,0.5], the model achieves the best performance, and under which conditions the model exhibits superior capability in addressing class imbalance issues between positive and negative classes.

E. Convergence Study

The convergence behavior of a model directly governs its training efficiency and resource consumption in practical deployment. This work evaluates the training stability of our model through systematic monitoring of loss trajectories and AP metrics during optimization across Corel5k and Pascal07 datasets. Experimental results demonstrate that on Corel5k, our ERF converges and gradually stabilizes around the 90-th epoch. In contrast, on Pascal07, ERF attains its peak performance near the 150-th epoch. Concurrently, the continuous improvement in AP further confirms the effectiveness and reliability of our model.

F. Complexity Analysis

The core computational overheads of the proposed model stem from four key components: the feature extraction module, the uncertainty-guided fusion module via evidence theory, the adaptive negative label pseudo-labeling, and the feature reconstruction module. Specifically, let $d_{\max}$ be the maximum neuron count of the autoencoder, the computational complexity of the feature extraction module is about $O(nmd_{\max}^2)$. The computational complexity of the other three modules are about $O(nmc)$, $O(nc)$, and $O(nmd_{\max}^2)$, respectively. Thus, the overall computational complexity of our ERF is about $O(nmd_{\max}^2 + nmc + nc)$.

In addition, we carry out an exhaustive runtime comparison across 10 methods in both the training and testing stages on the Corel5k dataset, where 70% samples are selected as training data. Relevant experimental outcomes are summarized in Table VI. This table shows that ERF takes 213.97 units of time at the training phase. Although this is higher than lightweight models like LVSL, it is significantly lower than models like RANK and DIC-net. At the inference phase, ERF takes 0.35 units of time, which is faster than models like MTD and LVSL, and only slightly higher than the optimal inference models like NAIML, demonstrating competitive inference efficiency.

V. CONCLUSION

In this paper, we present a novel method: Evidential Reliable Fusion (ERF) for Partial Multi-view Incomplete Multi-label Classification. The method provides two main contributions: 1) For multi-view fusion, we propose an uncertainty-guided fusion module via evidence theory. This module pioneers the integration of evidence theory at the early feature fusion stage, establishing a fine-grained fusion paradigm. 2) For incomplete multi-label classification, we propose an adaptive negative label pseudo-labeling strategy. By leveraging the model's perception ability for neighboring samples in the label space, we estimate its inference capability for missing label positions. Ultimately, we filter out reliable negative pseudo-labels to augment supervisory signals during training. The uncertainty-guided fusion module provides a reliable feature basis for the adaptive negative label pseudo-labeling strategy. The adaptive negative pseudo-labeling strategy compensates for missing supervisory signals by recovering reliable negative labels, thereby optimizing model performance.

Despite its strong performance in partial multi-view incomplete multi-label classification, the ERF framework still has limitations: its evidence-theory-based parameters are manually set without adaptivity. It fails to explicitly model label correlations, which are critical for generating high-quality pseudo-labels. Future work will be dedicated to overcoming these limitations.

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